1) NumPy-based implementation is significantly faster and more efficient than pure Python loops for large datasets because:

1- **Vectorization:** NumPy operations are vectorized, meaning they operate on whole arrays at once instead of iterating element by element. This eliminates Python-level loops.
2- **Optimized C backend:** NumPy is implemented in C under the hood, so the heavy computations run at compiled code speed, not interpreted Python speed.
3- **Memory efficiency:** NumPy arrays store data in contiguous memory blocks with fixed data types, reducing memory overhead compared to Python lists.
4- **Reduced Python overhead:** Loops in Python involve interpreter overhead for each iteration; NumPy minimizes this by performing operations at a lower level.

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2)When features are highly correlated (multicollinearity), the Normal Equation

$$\theta = (X^T X)^{-1} X^T y$$

can fail because $X^T X$becomes singular or nearly singular, meaning it cannot be inverted. This happens because correlated features make some columns of $X$almost linearly dependent, causing loss of independent information.

**Solution without removing features:** use regularization methods like Ridge Regression (L2 regularization), which adds a small value to the diagonal of $X^T X$to make it invertible and stabilize the solution.